



Modelling Financial Markets

Chapter 14: Deep Learning and Time-Series Foundation Models



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Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- ACI: Adaptive Conformal Inference
- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- BET: Bucharest Exchange Trading (index)
- CI: Confidence Interval
- COVID-19: COronaVirus Disease 2019
- CPU: Central Processing Unit
- CRPS: Continuous Ranked Probability Score
- CRSP: Center for Research in Security Prices
- DM: Diebold–Mariano (test)
- DQ: Dynamic Quantile (test)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- FHS: Filtered Historical Simulation
- FRED: Federal Reserve Economic Data
- FZ0: Fissler–Ziegel loss, zero-homogeneous version
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GIFT: General Time Series Forecasting Model Evaluation (benchmark GIFT-Eval)
- GPD: Generalised Pareto Distribution
- GRU: Gated Recurrent Unit
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- HS: Historical Simulation
- LASSO: Least Absolute Shrinkage and Selection Operator

Acronyms used in this material (2/2)

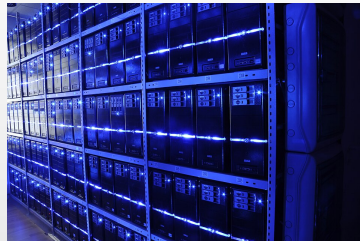
- LLM: Large Language Model
- LR: Likelihood Ratio (test)
- LSTM: Long Short-Term Memory (network)
- MCS: Model Confidence Set
- MDE: Minimum Detectable Effect
- MLP: MultiLayer Perceptron
- MSE: Mean Squared Error
- NBER: National Bureau of Economic Research
- PIT: Probability Integral Transform
- QLIKE: Quasi-LIKElihood (loss function)
- QS: Quantile Score
- RFF: Random Fourier Features
- RNN: Recurrent Neural Network
- RV: Realised Variance
- SPA: Superior Predictive Ability (test)
- TS: Time Series (in the name TS-Arena)
- US: United States
- VAR: Vector AutoRegression

Chapter Route and Outcomes

- Question: do deep networks and time-series **foundation models** forecast **returns**, **volatility** and **tail risk** better than the classical models of Chapters 5–9?
 - ▶ **Deep learning**: models built from many layers of simple non-linear units, trained on data; since 2024, foundation models pretrained on billions of observations
 - ▶ Answered with the evaluation tools of Chapter 8: strong benchmarks, loss differentials, backtests
- Route over two sessions
 - ▶ Session 1: networks for sequences, attention, foundation models, Test 1 (returns)
 - ▶ Session 2: Test 2 (realised volatility), Test 3 (VaR 1% and ES 2.5%), case study, AI for scientific discovery
- By the end you can
 - ▶ explain vanishing gradients as VAR impulse responses and the LSTM cell state as an AR(1)
 - ▶ turn a zero-shot quantile grid into point, volatility and VaR/ES forecasts
 - ▶ test a new model correctly: nesting, multiplicity, calibration, contamination and power

Three Forecasting Targets

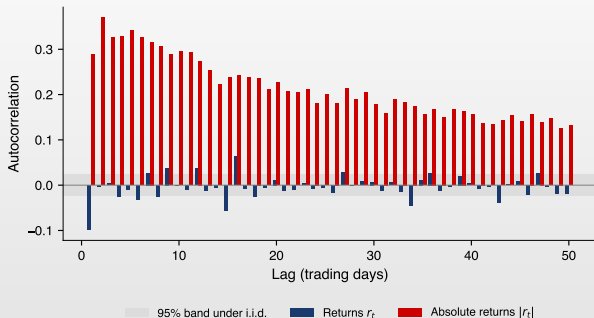
- ▣ **Returns** r_t (conditional mean)
 - ▶ Very low signal-to-noise ratio; predictable patterns are traded away (Chapter 2)
 - ▶ Benchmark: the zero forecast
- ▣ **Volatility** (conditional variance)
 - ▶ Persistent and clustered (Chapters 1 and 5): there is a signal to learn
 - ▶ Benchmark: log-HAR
- ▣ **Tail-risk measures**: VaR 1% and ES 2.5%
 - ▶ Need the shape of the lower tail, not just the centre of the distribution (Chapter 7)
 - ▶ Benchmark: FHS



A data centre: deep networks are trained on racks of such servers

Photo: [BalticServers.com](#) (2013); CC BY-SA 3.0; Wikimedia Commons

Where Is the Signal?



- S&P 500 daily returns since 2000 (6 718 days), lags 1–50; grey band: $\pm 1.96/\sqrt{T} = \pm 0.024$ under independence
- Returns: lag-1 autocorrelation -0.099; 15 of 50 lags outside the band, small and of changing sign: little to learn about the mean
- Absolute returns: 0.29 at lag 1, still 0.13 at lag 50; all 50 lags outside the band: persistent volatility clustering
- Prediction for the chapter: flexible models can find the volatility signal, not much in the mean

From the Multilayer Perceptron to Deep Learning

- MLP (multilayer perceptron, Chapter 13): $h^{(\ell)} = \varphi \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right)$,
 $h^{(0)} = x$; φ : ReLU, tanh or sigmoid $\sigma(z) = 1/(1 + e^{-z})$
 - ▶ Backpropagation (Rumelhart, Hinton & Williams (1986)) and Adam (Kingma & Ba (2015)); loss MSE for points, pinball $(u - 1\{y < q\})(y - q)$ for a quantile of level u
- Universal approximation (Hornik, Stinchcombe & White (1989)): silent on the sample size needed, the binding constraint with one financial series
 - ▶ Regularisation: early stopping, weight decay, ensembles over seeds; the validation block must follow the training block
- 2024 Nobel Prize in Physics to Hopfield and Hinton (LeCun, Bengio & Hinton (2015) for the methods); in finance, Gu, Kelly & Xiu (2020) gain on a cross-section of thousands of stocks, not on one index



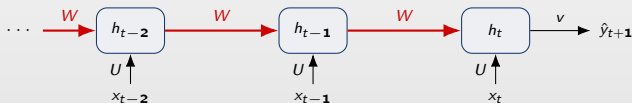
Geoffrey Hinton, Nobel laureate 2024

Photo: Arthur Petron (2024); CC BY-SA 4.0;

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RNN and LSTM in One Slide

- ▣ **Recurrent neural network (RNN):** $h_t = \tanh(W h_{t-1} + U x_t + b)$, $\hat{y}_{t+1} = v^\top h_t + c$
 - ▶ The same weights at every step: unrolled over time, a very deep network with shared weights
- ▣ **Long short-term memory (LSTM, Hochreiter & Schmidhuber (1997)):** gates $f_t, i_t, o_t = \sigma(W_\bullet x_t + U_\bullet h_{t-1} + b_\bullet)$, candidate $g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g)$
 - ▶ Cell state $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, output $h_t = o_t \odot \tanh(c_t)$; $\odot$: element-by-element product
 - ▶ f_t : forget gate, i_t : input gate, o_t : output gate; GRU (gated recurrent unit): a lighter variant with two gates



Vanishing Gradients: a Bound and a VAR(1)

- Backpropagation through time multiplies one Jacobian per step

▶ $\frac{\partial h_T}{\partial h_{T-k}} = \prod_{j=0}^{k-1} D_{T-j} W, \quad D_t = \text{diag}(\varphi'(Wh_{t-1} + Ux_t + b))$

- ▶ Submultiplicativity: $\|\partial h_T / \partial h_{T-k}\|_2 \leq (\gamma_\varphi \|W\|_2)^k$, $\gamma_\varphi = \sup_z |\varphi'(z)|$ ($= 1$ for $\tanh$): $\gamma_\varphi \|W\|_2 < 1$ suffices for vanishing (Pascanu, Mikolov & Bengio (2013))

- Linearise at a fixed point h^* : $h_t - h^* \approx A(h_{t-1} - h^*) + DUx_t$, $A = DW$: a VAR(1) (vector autoregression) in the hidden state

- ▶ $\partial h_T / \partial h_{T-k} = A^k$ is its impulse response at horizon k : stable VAR, $\rho(A) < 1$, gradients decay like $\rho(A)^k$; $\rho(A) > 1$, they explode
- ▶ A stable RNN near its fixed point is a short-memory model, like any stationary VAR (Seminar A1; details in the Appendix)

- Scalar case $|w|^k$: $0.9^{20} = 0.12$, $0.9^{50} = 0.005$ (Bengio, Simard & Frasconi (1994))

- ▶ Remedies: clipping the gradient norm (explosion); gating, as in the LSTM (vanishing)

Why the LSTM Remembers

- Along the cell state the only multiplier is the forget gate
 - ▶ $\frac{\partial c_T}{\partial c_{T-k}} = \prod_{j=0}^{k-1} f_{T-j}$ (direct path); $c_t = f_t \odot c_{t-1} + i_t \odot g_t$: a random-coefficient AR(1)
- Constant forget gate $f = \sigma(b_f)$: an AR(1) with coefficient f
 - ▶ $b_f = 0$: $f = 0.5$, $0.5^{50} \approx 9 \times 10^{-16}$
 - ▶ $b_f = 3$: $f = 0.953$, $0.953^{50} \approx 0.09$; half-life 14 steps
- Short memory by construction: $\text{Corr}(c_t, c_{t-k}) = f^k$ decays geometrically; long memory decays hyperbolically, $\rho_k \sim Ck^{2d-1}$
 - ▶ Fitted to the S&P 500 absolute returns, $f = 0.984$ predicts 0.027 at lag 150 against 0.059 observed (Appendix)

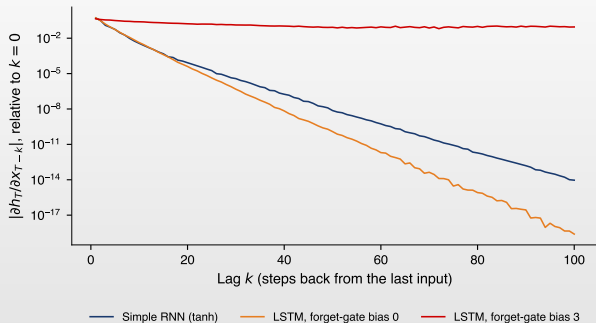


Sepp Hochreiter, co-inventor of the LSTM

Photo: Eulenreich (2015); CC BY-SA 4.0;

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The Vanishing Gradient, Measured

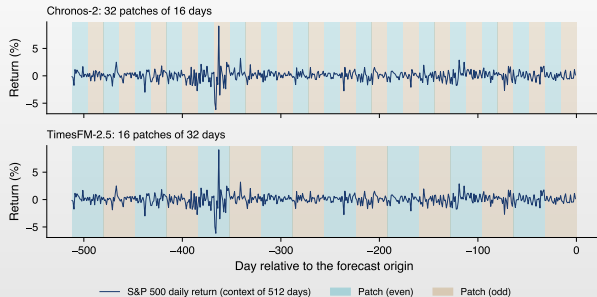


- $|\partial h_T / \partial x_{T-k}|$ by automatic differentiation; 16 hidden units, 50 random initialisations, same Normal inputs
- Simple RNN: 8.3×10^{-5} at lag 20, 7.4×10^{-9} at lag 50: nothing is learned from inputs more than a few weeks back
- LSTM, forget-gate bias 0: 1.2×10^{-10} at lag 50, no better than the RNN ($f \approx 0.5$ halves the memory at every step)
- LSTM, bias 3: 8.4% at lag 50, as $0.953^{50} \approx 0.09$ predicts; initialise the forget-gate bias at a positive value

Attention, Positions and Patches

- Each time step asks a question (query) and compares it with every past step (key):
$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^{\top} / \sqrt{d_k}\right) V \text{ (Vaswani et al. (2017))}$$
 - ▶ $Q = XW_Q$, $K = XW_K$, $V = XW_V$: linear projections of the inputs X ; softmax turns scores into weights summing to one; $\sqrt{d_k}$ keeps it from saturating
 - ▶ No product of Jacobians over time: any past step is one hop away; cost $O(n^2)$ in the number of inputs n
- **Positional encoding**: attention treats its inputs as a set, so order is added (sinusoids in the original Transformer; rotary embeddings in Chronos-2)
- **Patching** (Nie et al. (2023)): a token is a block of consecutive observations
 - ▶ 512 daily returns in patches of 16 days: 32 tokens, about 256 times fewer attention scores; each token summarises a few weeks of market activity

Patching a 512-Day Context



- The last 512 daily returns of the S&P 500, split as Chronos-2 (16 days) and TimesFM-2.5 (32 days) read them
- Each shaded block becomes one embedding vector for the Transformer
- DeepAR (Salinas et al. (2020)), N-BEATS (Oreshkin et al. (2020)) and PatchTST (Nie et al. (2023)) learn across many series; foundation models push this to its limit

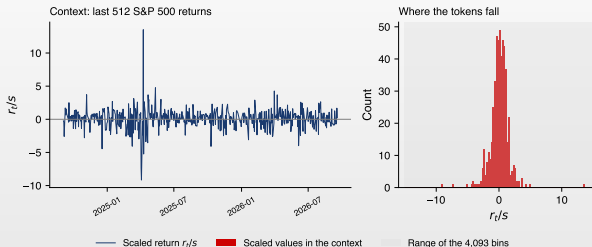
What Is a Foundation Model?

- ▣ A large network pretrained once on a very broad corpus, then used on new tasks
 - ▶ Idea borrowed from large language models (LLM): pretrain on text, then use on any question
- ▣ **Zero-shot** forecasting: no training on the target series; only its recent history (the **context**) is passed at forecast time
 - ▶ Output: a set of quantiles of the future values, not only a point
 - ▶ Fine-tuning (further training on the target series) is possible but not used here
- ▣ Appeal for finance: one model for thousands of series, no estimation per series

Chronos: Time Series as a Language

- ▣ [Ansari et al. \(2024\)](#): turn values into tokens and train a language model (T5) with cross-entropy
 - ▶ Mean scaling: divide the context by $s = \frac{1}{C} \sum_t |x_t|$
 - ▶ Quantisation: 4,093 equally spaced bin centres in $[-15, 15]$ (width 0.0073), plus special tokens
 - ▶ Forecast: sample future tokens, map back to values, read quantiles from the samples
- ▣ Pretraining corpus: public datasets (energy, transport, retail, weather, web, finance and economics) plus synthetic series (TSMixup, KernelSynth)
 - ▶ Model sizes 20 to 710 million parameters; the small version has 46 million

What Chronos Sees: Scaled Returns and Tokens



- Context: the last 512 S&P 500 daily returns; scale $s = 0.672\%$; scaled returns from -9.2 to 13.5, 96% within ± 3
- The 512 returns use 336 distinct tokens out of 4,093: scaling makes the model free of units, so the volatility level must be read from the context
- A crash day M raises s by about $|M|/C$: strongly in a short context, little in a 512-day one (Seminar, A3)

The Three Models We Test

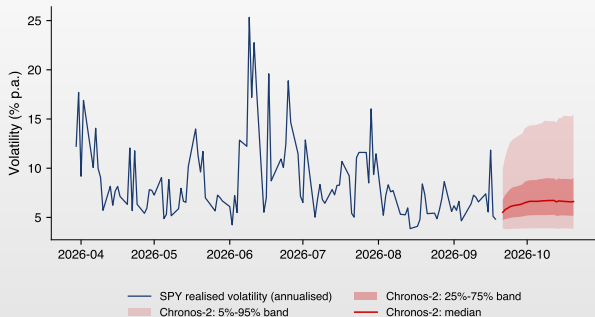
Model	Parameters	Patch	Quantile levels	Weights (commit)
Chronos-Bolt (small)	48 M	16	10%–90%	25.11.2024
TimesFM-2.5	231 M	32	mean, 10%–90%	01.10.2025
Chronos-2	119 M	16	1%, 5%, . . . , 99%	30.10.2025

- **Chronos-Bolt:** patches and a direct quantile output instead of token sampling
- **Chronos-2** ([Ansari et al. \(2025\)](#)): robust scaling, attention across time and across groups of related series
- **TimesFM** ([Das et al. \(2024\)](#)): decoder-only Transformer, pretrained on about 10^{11} time points
- All run on a CPU with a 512-day context; weight dates bound what the models could have seen; only Chronos-2 reaches the 1% level

From a Quantile Grid to VaR and ES

- The model returns quantiles q_u at levels $u_1 < \dots < u_K$
 - ▶ Point forecast: the median $q_{0.5}$, or the mean $\int_0^1 q_u du$; for increasing g , $E[g(Y)] = \int_0^1 g(q_u) du$ (e.g. $g = \exp$ for log variance)
 - ▶ Tail risk (Chapter 7): $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
- Worked example, S&P 500, day after 18.09.2026: $q_{0.01} = -2.70\%$, $q_{0.05} = -1.40\%$
 - ▶ VaR 1% = 2.70%; linear interpolation $q_{0.025} = q_{0.01} + \frac{0.015}{0.04} (q_{0.05} - q_{0.01}) = -2.21\%$, VaR 2.5% = 2.21%
 - ▶ ES 2.5% = $-\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.56\%$
- Below the lowest level the grid is silent: holding q_u at $q_{0.01}$ understates ES, interpolation can err either way (Seminar, A5–A6)
 - ▶ Chronos-Bolt and TimesFM stop at 10%: a "VaR 1%" request returns the 10% quantile

A Zero-Shot Fan Chart for Volatility



- Chronos-2 on log realised variance of SPY (512-day context), 22 trading days ahead from 18.09.2026; bands: quantiles 5–95% and 25–75%
- Last volatility 4.8% p.a., 20-day average 6.6%; median forecast at 22 days 6.6%, 90% band 3.8–15.4%
- The band widens and is skewed upwards: mean reversion, as in GARCH and HAR (Chapters 5, 9), with no estimation on SPY
- Whether these forecasts are better is the question of Tests 1–3

 MFM_ch14_fan_chart

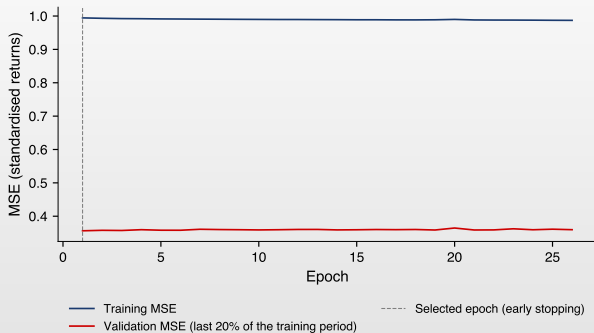
Pretraining Leakage and Contamination

- A test is out of sample only if the test period was never seen during training; four ways it can be “seen”
 - ▶ (i) the same series is in the pretraining corpus; (ii) correlated series carry the same shocks; (iii) developers select checkpoints on overlapping public benchmarks
 - ▶ (iv) our own look-ahead: scalings, windows, seeds or contexts chosen with test data
- The Chronos corpus includes financial series (e.g. daily exchange rates 1990–2016); we cannot rule out that our series were seen
- Remedy for (i)–(iii): a **post-release window** from 3 November 2025, a common cutoff after all three sets of weights were published
 - ▶ Clean but short (about 220 trading days; 320 calendar days for Bitcoin): low power, especially for VaR 1%
- (iv) is under our control: every scaling, window and estimate uses only past data; a formal protocol is in the Appendix

Design of the Return Test

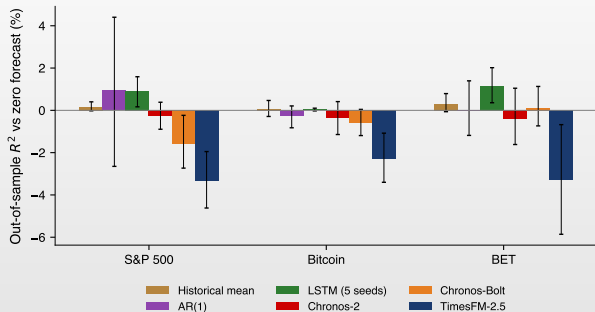
- Daily log returns (%): S&P 500 and BET from 04.01.2016 (2 693 and 2 675 days), Bitcoin from 01.01.2019 (2 818 days)
 - ▶ Every forecast uses only data up to the previous close
- Models: historical mean, AR(1) on a 1,000-day window, LSTM, Chronos-2, Chronos-Bolt and TimesFM-2.5 (median)
 - ▶ Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 -0.77% (Chronos-2), -3.42% (TimesFM-2.5)
- Benchmark: the zero forecast; out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$
 - ▶ Zero rather than the historical mean, as in [Gu, Kelly & Xiu \(2020\)](#); compare [Campbell & Thompson \(2008\)](#)
 - ▶ Inference: block-bootstrap CI for R^2 (blocks of 20 days), Diebold–Mariano (DM) test on squared errors with Newey–West variance ([Diebold & Mariano \(1995\)](#))

The LSTM for Returns and Early Stopping



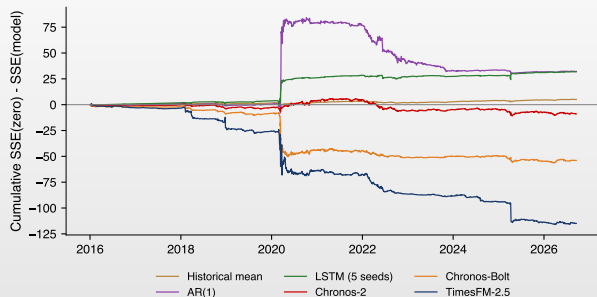
- Input: the last 20 returns, standardised on the training period; 16 hidden units, linear output: $4(hd + h^2 + 2h) + h + 1 = 1233$ parameters
- MSE, Adam, weight decay 10^{-4} ; trained once on up to 4,000 days, last 20% for early stopping; forecast = average of 5 seeds
- Chart (S&P 500, seed 0): training stops 26 epochs in, the weights of epoch 1 are kept
- Little to learn: forecast s.d. 0.047% against 1.13% for returns; validation MSE is lower only because those years were calmer

Out-of-Sample R^2 for Returns



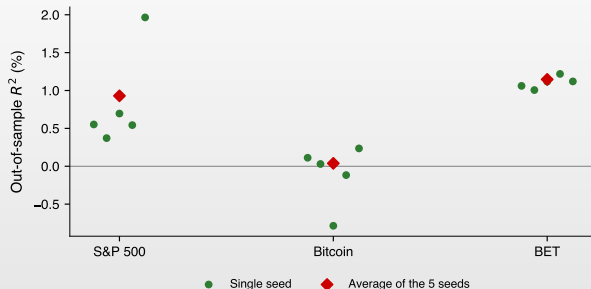
- R^2 against the zero forecast (%), 95% block-bootstrap CIs; full table with DM p -values in the Appendix
- S&P 500 and Bitcoin: no model beats zero significantly
- TimesFM-2.5 is significantly **worse** than zero (S&P 500 R^2 -3.35%, DM $p = 0.001$): forecast s.d. 0.141%, three times the LSTM's
- BET: LSTM R^2 +1.15% (DM $p = 0.014$): a small signal, tested for nesting, multiplicity and stale prices below

Cumulative Squared-Error Gains on the S&P 500



- $\sum_{s \leq t} [r_s^2 - (r_s - \hat{r}_s)^2]$: rises when the model beats the zero forecast, falls when it loses
- All curves stay near zero in calm years and jump in March 2020: one month decides the ranking
- AR(1) gains in 2020 from the negative autocorrelation of crash days, then gives part back; TimesFM-2.5 loses steadily
- Report the path, not only the final R^2

The Seed Lottery



- Out-of-sample R^2 of five LSTMs that differ only in the random seed, and of their average
- Bitcoin: single seeds from -0.79% to +0.24%; S&P 500 from +0.37% to +1.96%: variation as large as the effect
- BET: all 5 seeds between +1.01% and +1.22%: robust across seeds, not yet robust to multiplicity or stale prices
- Choosing the best seed after seeing the test is backtest overfitting (Chapter 13)

Nested Models and Many Tests

- Zero is nested in the historical mean and AR(1): under H_0 the larger model only adds estimation noise, so DM rejects too rarely
 - ▶ Clark & West (2007): $\tilde{d}_t = e_{0,t}^2 - [e_{1,t}^2 - (\hat{r}_{0,t} - \hat{r}_{1,t})^2]$, one-sided HAC t -test
 - ▶ One-sided p , DM vs Clark–West: historical mean 0.048 vs 0.007 (S&P 500), 0.041 vs 0.003 (BET); AR(1) 0.320 vs 0.055
- 6 models $\times$ 3 markets = 18 tests; the smallest one-sided DM p is the BET LSTM, 0.007
 - ▶ Bonferroni threshold 0.0028; Holm-adjusted p over the 18 tests 0.130: not significant
 - ▶ Within the six BET models: Romano–Wolf (Romano & Wolf (2005)) 0.016; SPA (Hansen (2005)), H_0 : no model beats zero: BET 0.018, S&P 500 0.130, Bitcoin 0.354
- Reading: “some model beats zero on the BET” survives; “one of 18 comparisons is real” does not. Fix the family before looking

Is BET Predictability Real? Non-Synchronous Trading

- Stale prices of thinly traded constituents make index returns positively autocorrelated, even if every price is efficient (Lo & MacKinlay (1990))
 - ▶ First-order autocorrelation since 2016: BET 0.06, S&P 500 -0.14
 - ▶ The LSTM forecast has correlation 0.88 with the previous day's BET return: it mostly learned this autocorrelation
- Tradable check: the BET-tracking ETF Patria-TVBETETF, 2629 common days, same LSTM signal
 - ▶ Same day: R^2 2.88% (BET 1.26%), but the ETF is thinly traded too (autocorrelation 0.07, 4.2% zero-return days)
 - ▶ Executed one day later: R^2 0.57%, Clark–West $p = 0.003$
- Reading: the evidence is consistent with stale-price effects; a small signal survives the execution lag, before transaction costs (Chapter 2)

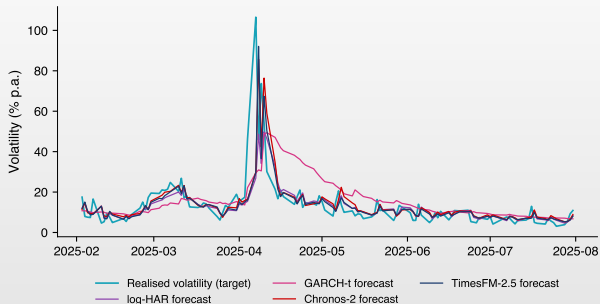
Target, Benchmark and Loss

- ▣ **Realised variance** (RV, Andersen et al. (2003)): sum of squared 5-minute log returns within the trading day
 - ▶ $RV_t = \sum_{j=1}^{78} r_{t,j}^2$ for SPY (regular session, 6.5 hours); average volatility 11.9% p.a.
- ▣ **HAR** (Corsi (2009)): $RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-5:t-1} + \beta_m \overline{RV}_{t-22:t-1} + \varepsilon_t$
 - ▶ log-HAR: the same on log RV, forecast $\exp(\hat{y} + \hat{s}^2/2)$: the mean if the log error is conditionally Normal
- ▣ **QLIKE** loss (Patton (2011)): $L = RV_t / \hat{h}_t - \log(RV_t / \hat{h}_t) - 1$
 - ▶ Minimised by the conditional mean; ranks forecasts as the true variance would when RV is a conditionally unbiased proxy, $E[RV_t | \mathcal{F}_{t-1}] = \sigma_t^2$ (so does MSE)

Design of the Volatility Test

- ▣ SPY realised variance, one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days)
 - ▶ HAR and log-HAR on a rolling 500-day window; random walk: $\hat{h}_t = \text{RV}_{t-1}$
 - ▶ GARCH-t on daily SPY returns, rescaled to RV units by the ratio of average RV to average squared return over 500 days
 - ▶ LSTM on the last 22 values of log RV, retrained every 125 days on 500 days (hidden size 8, three seeds)
- ▣ Foundation models on log RV (context 512): forecast $\hat{h}_t = \int_0^1 \exp(q_u) du$ on the quantile grid
 - ▶ Post-release window: 218 days from 3 November 2025

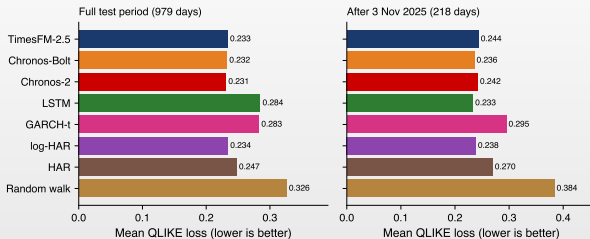
Forecasts Through the April 2025 Shock



- SPY realised volatility (annualised) and one-day-ahead forecasts, February–July 2025; peak 107% p.a. on 07.04.2025, after the US tariff announcements
- No model anticipates the jump; all react the next day: one-day forecasts extrapolate the past, shocks come from news
- Foundation models and log-HAR decay back at similar speeds; GARCH-t reacts to daily returns and decays more slowly

MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models



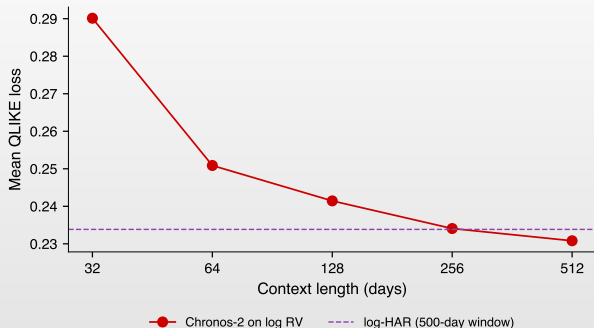
- Left: full test period; right: after 3 November 2025 (218 days); full table with DM and MCS p -values in the Appendix
- Chronos-2 (0.231), Chronos-Bolt, TimesFM-2.5 and log-HAR (0.234): DM tests not significant (Chronos-2 $p = 0.537$), all in the 90% MCS
- Random walk and GARCH-t are eliminated; HAR in levels over-forecasts (mean forecast / mean RV = 1.36); after release the small LSTM is best (0.233): rankings are unstable
- No conditional difference: Giacomini-White test ($1, d_{t-1}$) against log-HAR, $p = 0.760$ (Chronos-2), 0.935 (TimesFM-2.5); no difference detected is not proof of equality

Median or Mean? A Costly Detail

Chronos-2 and TimesFM-2.5 on log RV	QLIKE	Mean forecast / mean RV
Chronos-2, $\int_0^1 \exp(q_u) du$	0.231	1.02
Chronos-2, $\exp(q_{0.5})$	0.259	0.79
TimesFM-2.5, $\int_0^1 \exp(q_u) du$	0.233	0.94
TimesFM-2.5, $\exp(q_{0.5})$	0.259	0.79

- A theorem, not a curiosity ([Gneiting \(2011\)](#)): QLIKE is consistent for the mean, $\arg \min_h E[L(RV, h) | \mathcal{F}] = E[RV | \mathcal{F}]$
 - ▶ Monotone maps preserve quantiles: $\exp(\text{median of } \log RV)$ is the median of RV, elicited by absolute error, not by QLIKE
 - ▶ Lognormal RV: median/mean = $e^{-s^2/2}$; the median version is significantly worse (DM +3.60 for Chronos-2) and worse than log-HAR (0.234)

How Much Context Does Chronos-2 Need?

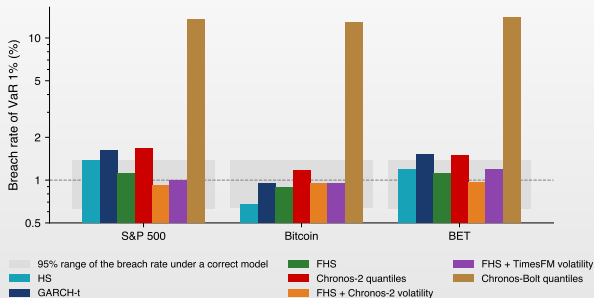


- QLIKE of Chronos-2 on log RV with contexts of 32 to 512 days, on the same forecast days
- QLIKE falls from 0.290 (32 days) to 0.231 (512 days); below about 256 days Chronos-2 is worse than log-HAR (0.234)
- Long memory: months of history reveal the slow-moving level, which HAR encodes explicitly through its 22-day average
- Context length is a modelling choice to report and test, like a window length

Design of the Tail-Risk Test and the FZ0 Loss

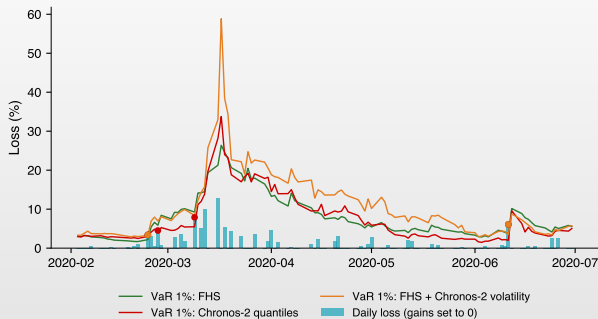
- $\text{VaR}_\alpha = -q_\alpha(r_{t+1} \mid \mathcal{F}_t)$, the loss exceeded with probability α ; $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$; levels VaR 1% (coverage tests) and (VaR 2.5%, ES 2.5%)
- Classical models, 1,000-day window, re-estimated every 20 days: HS, GARCH(1,1)-t, FHS (McNeil & Frey (2000))
 - ▶ **Raw**: Chronos-2 quantiles of tomorrow's return (context: 512 returns)
 - ▶ **Hybrid**: scale $\hat{\sigma}_t$ = forecast of $|r_t|$ (Chronos-2: grid mean; TimesFM: median); FHS on $r_t/\hat{\sigma}_t$
- ES alone is not elicitable, the pair (VaR, ES) is (Fissler & Ziegel (2016)); FZ0 (Patton, Ziegel & Chen (2019)), loss $L_t = -r_t$, $v = \text{VaR}$, $e = \text{ES} > 0$:
 - ▶ $\text{FZ0} = \frac{1\{L_t > v\}(L_t - v)}{\alpha e} + \frac{v}{e} + \log e - 1$
 - ▶ Lower is better: penalises breaches (first term) and wide forecasts ($\log e$); models compared by DM tests on FZ0 differences and by the MCS (Hansen, Lunde & Nason (2011))

VaR 1% Breach Rates



- Share of days with a loss above VaR 1% (log scale); grey band: 95% range if the model were correct
- Raw Chronos-2: too many breaches on the S&P 500 (1.67%, Kupiec $p = 0.001$) and the BET (1.50%)
- Chronos-Bolt, lowest quantile 10% used as "VaR 1%": 13.5%
- Hybrids: breach rates close to 1% in all three markets (average coverage only; full table in the Appendix)

The COVID-19 Crash: VaR 1% of Three Models



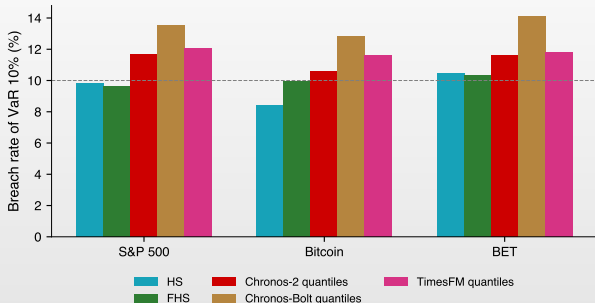
- S&P 500, February–June 2020 (104 days): daily losses and VaR 1% forecasts; dots: breaches
- Breaches: FHS 2, raw Chronos-2 4, FHS with Chronos-2 volatility 2, HS 12; all models are hit in the first days
- Chronos-2 reads the context fast, so its quantiles widen quickly
- The hybrid overshoots: VaR 1% of 58.9% on 17.03.2020, against 26.4% for FHS and a largest loss of 12.8% (Appendix)

Backtesting Beyond Kupiec: Clustering and the DQ Test

p : indep. / DQ	HS	GARCH-t	FHS	Chronos-2 raw	FHS + Chronos-2	FHS + TimesFM
S&P 500	0.001 / <0.001	0.205 / 0.001	0.045 / 0.021	0.223 / <0.001	0.021 / 0.001	0.029 / <0.001
Bitcoin	0.611 / 0.760	0.261 / 0.463	0.220 / 0.092	0.404 / 0.034	0.026 / 0.004	0.261 / 0.182
BET	0.399 / <0.001	0.258 / 0.082	0.409 / 0.650	0.270 / 0.026	0.475 / <0.001	0.379 / <0.001

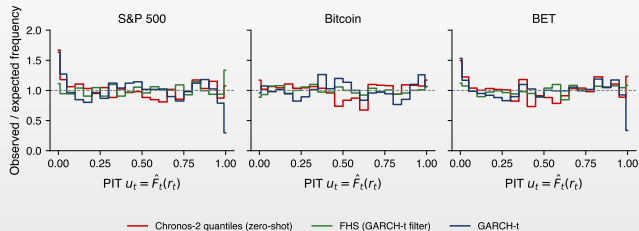
- Independence ([Christoffersen \(1998\)](#)): first-order Markov chain of breaches, $LR \sim \chi^2_1$; conditional coverage = Kupiec + independence, χ^2_2
- Dynamic quantile (DQ, [Engle & Manganelli \(2004\)](#)): regress $I_t - \alpha$ on a constant, four lagged hits and VaR_t ; Wald statistic $\sim \chi^2_6$
- S&P 500: the hybrids have the right breach rate but fail independence and DQ; every model fails DQ at 5%
 - BET: FHS passes both tests, the hybrids fail DQ; right on average is not right conditionally

Where Deciles Suffice: VaR 10%



- Breach rate of the 10% quantile (target 10%) for HS, FHS and the three foundation models
- S&P 500: HS 9.8%, FHS 9.6%; Chronos-2 11.7%, Chronos-Bolt 13.5%, TimesFM-2.5 12.0%
- Kupiec rejects the foundation models in eight of nine cases: the lower tail sits too high, through location, scale or tail shape
- Plausible reason: pretraining corpora are dominated by smoother series; as volatility filters in FHS the models inherit the tails from the data

Calibration of the Whole Distribution: the PIT



- PIT (probability integral transform) $u_t = \hat{F}_t(r_t)$: i.i.d. uniform if the predictive distribution is correct (Diebold, Gunther & Tay (1998)); bars: observed over expected frequency in the 22 intervals of the Chronos-2 grid
- Berkowitz test (Berkowitz (2001)) of $\mu = 0, \sigma = 1$ for $\Phi^{-1}(u_t)$, censored outside [1%, 99%]: Chronos-2 rejected on the S&P 500 ($p = 0.019, \hat{\sigma} = 1.03$) and the BET; FHS never
- Quantile score $QS = \frac{2}{K} \sum_k (1\{r_t < q_{u_k}\} - u_k)(q_{u_k} - r_t)$, a grid analogue of the CRPS (Gneiting & Raftery (2007)): Chronos-2 worse than FHS in the centre and the tails (S&P 500 DM +2.51, +3.65); table in the Appendix

Comparing (VaR 2.5%, ES 2.5%) Forecasts: the MCS

HS	0.140	0.343	0.015
GARCH-t	0.187	0.597	0.341
FHS	1.000	0.597	1.000
Chronos-2 quantiles	0.140	0.641	0.239
FHS + Chronos-2 volatility	0.140	1.000	0.341
FHS + TimesFM volatility	0.183	0.704	0.202
	S&P 500	Bitcoin	BET

- p -values of the 90% model confidence set (Hansen, Lunde & Nason (2011)) for the FZ0 loss; green: kept, red: eliminated; FZ0 table in the Appendix
- S&P 500: FHS has the lowest FZ0; the hybrids have the right breach rate but clustered breaches and a larger VaR (3.20% vs 2.78%)
- The MCS eliminates only HS on the BET; pairwise DM is sharper: FHS beats the Chronos-2 hybrid on the S&P 500 (DM +2.65, $p = 0.008$) and the BET (DM +2.36)

After the Release: Few Breaches, Little Power

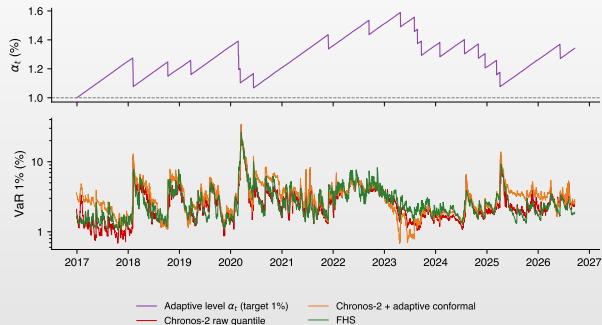
Breaches after 3.11.2025	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (220 days)	0	2	2	2	2	2
Bitcoin (320 days)	3	3	4	4	3	4
BET (217 days)	4	4	4	5	1	2

- Expected breaches: 2.2 (S&P 500), 3.2 (Bitcoin), 2.2 (BET): no contamination is possible, but the tests have little power
 - ▶ Exact two-sided Kupiec test at 5% against a true rate of 1.67%: power 0.19 at $T = 220$, actual size 0.13; 0.50 at $T = 1,000$; 0.87 at $T = 2,693$
 - ▶ 80% power first at about 2 130 days (8.4 years), for good from 2 320; against 2%: power 0.29 at $T = 220$ (Seminar A8–A9)
- Low power is not zero power: HS has 0 breaches on the S&P 500, and $x = 0$ is in the Kupiec rejection region at $T = 220$; a non-rejection does not establish calibration

Recalibrating Quantiles: Adaptive Conformal Inference

- Score $E_t = \hat{q}_{\alpha,t} - r_t$; corrected quantile $\hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}$, with $\hat{Q}$ the empirical $(1 - \alpha_t)$ quantile of the last 250 scores
- Adaptive level (Gibbs & Candès (2021)): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, $\text{err}_t = 1\{r_t < \hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}\}$, $\alpha_1 = \alpha$, $\gamma = 0.001$;
 $\gamma = 0$: plain split conformal
 - After a breach α_t falls (the next VaR is wider); after a calm day it rises slowly
- Guarantee for any sequence, no model needed: $|\frac{1}{T} \sum_t \text{err}_t - \alpha| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$, a long-run statement only
- ES 2.5%: Chronos-2 ES shifted by the VaR 2.5% correction, a heuristic without a conformal guarantee; rules and the full table in the Appendix

Adaptive Conformal VaR 1% on the S&P 500



- Top: adaptive level α_t (target 1%); bottom: VaR 1% of raw Chronos-2, Chronos-2 with ACI (adaptive conformal inference), FHS (log scale)
- ACI brings the breaches near 1% (S&P 500: 21 vs 24.4 expected); conditional coverage is not rejected in any market
- DQ still rejects on the S&P 500 and the BET; the VaR is wider (3.36% vs 2.85% for FHS) and FZ0 significantly higher
- Proper scoring rules reward sharpness subject to calibration ([Gneiting & Raftery \(2007\)](#)): calibration is necessary, not sufficient

Case Study: Is Complexity a Virtue in Return Prediction?

- **Question** (Kelly, Malamud & Zhou (2024)): must a return-prediction model have far fewer parameters P than observations T , or can models with complexity $c = P/T > 1$ do better out of sample?
 - ▶ Setting (eqs. 1–2): $R_{t+1} = f(G_t) + \varepsilon_{t+1}$, f unknown, approximated by a wide one-hidden-layer network $\sum_{i=1}^P S_{i,t} \beta_i$ (universal approximation, Hornik, Stinchcombe & White (1989))
- **Link to Test 1**: on the S&P 500 and Bitcoin no model beats the zero forecast significantly; the paper asks whether a negative out-of-sample R^2 means no economic value
 - ▶ Economic value: Sharpe ratio of the market-timing return $\pi_t R_{t+1}$, with position $\pi_t = \hat{\beta}' S_t$ (eq. 6)
- **Theory** (Sections III–IV): under the assumptions of the paper, a misspecified model with optimally chosen ridge shrinkage has expected out-of-sample R^2 and timing Sharpe ratio rising with c
 - ▶ Near $c = 1$ (the interpolation boundary) least squares fits the training data exactly and its variance explodes; beyond it, performance recovers
- **Why it matters**: a theoretical case for the large models of this chapter, against the parsimony rule of classical econometrics

Case Study: Data and Method

- **Data** (Section V.A): monthly excess return of the CRSP value-weighted index; the 15 predictors of [Welch & Goyal \(2008\)](#) (14 variables and the lagged market return), 1926–2020
 - ▶ Returns scaled by their trailing 12-month volatility, predictors by an expanding-window standard deviation; 36 months of warm-up, so the sample starts in 1930
- **Random Fourier features** (eq. 20, [Rahimi & Recht \(2007\)](#)): $S_{i,t} = (\sin(\gamma \omega_i' G_t), \cos(\gamma \omega_i' G_t))'$, $\omega_i \sim N(0, I)$, $\gamma = 2$
 - ▶ A wide two-layer network whose first-layer weights ω_i are drawn at random and fixed; only the output weights β are estimated
- **Ridge** (Section II.A): $\hat{\beta}(z) = (zI + T^{-1} \sum_t S_t S_t')^{-1} T^{-1} \sum_t S_t R_{t+1}$, no intercept; $z \rightarrow 0^+$: the ridgeless (pseudo-inverse) solution
- **Design** (Section V.C): rolling windows of $T = 12, 60, 120$ months; P from 2 to 12 000; $\log_{10} z = -3, \dots, 3$; results averaged over 1 000 independent draws of the features
- **Evaluation**: out-of-sample $R^2 = 1 - \text{Var}(\text{error})/\text{Var}(R)$ (footnote 40; the theory divides the MSE by $E[R^2]$, the zero forecast of Test 1); annualised Sharpe ratio; alpha and information ratio (alpha over residual volatility) from a regression of the timing return on the market
 - ▶ Benchmark: the linear "kitchen sink" regression on all 15 predictors (Section V.D)

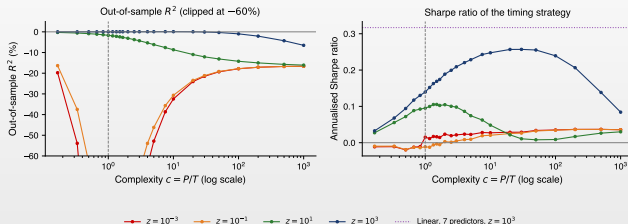
Case Study: Key Results

Model ($T = 12$)	R^2 (%)	Sharpe [t]	Inf. ratio [t]	Max loss	Skewness
Linear, $z = 0^+$	< -100	$-0.11 [-1.0]$	$-0.16 [-1.6]$	98.5	-0.9
Linear, $z = 10^3$	-3.8	$0.46 [4.4]$	$0.33 [3.1]$	2.4	-0.1
RFF, $c = 1000$, $z = 10^3$	0.6	$0.47 [4.5]$	$0.31 [2.9]$	1.2	2.5

Source: Table I, Panel A, of the paper; information ratio against the market; maximum monthly loss in standard deviations

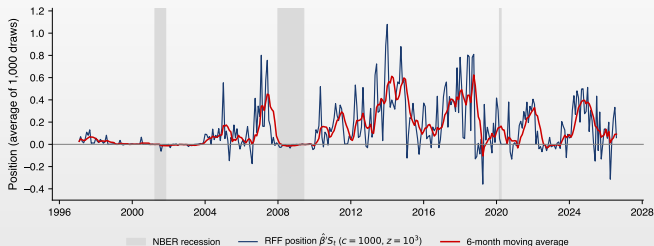
- ▣ Ridgeless kitchen sink ($P = 15 > T = 12$): R^2 below -100% , Sharpe ratio -0.11
 - ▶ Reading: at the interpolation boundary forecasts explode, reproducing the negative verdict of [Welch & Goyal \(2008\)](#)
- ▣ Same predictors with $z = 10^3$: $R^2 = -3.8\%$, yet Sharpe ratio 0.46 ($t = 4.4$)
 - ▶ Reading: R^2 penalises the scale of forecasts; timing profits depend on their correlation with returns (footnote 5)
- ▣ RFF, $c = 1000$: information ratio 0.26 ($t = 2.5$) against the best linear model; with $T = 60$ and 120 , 0.25 and 0.24 against the market ($t = 2.3$ and 2.2 ; Panels B–C)
 - ▶ Reading: nonlinearity extracts more from the same predictors, with a smaller maximum loss and positive skewness
- ▣ Positions (Figure 10): almost always long; cut before 14 of 15 NBER recessions, out of sample
 - ▶ Reading: the model learns the sign restriction of [Campbell & Thompson \(2008\)](#) without being given it

Case Study: Complexity on Course Data



- Figures 7–8 of Kelly, Malamud & Zhou (2024) on course data: SPY excess returns, 7 predictors, 354 test months (February 1997–August 2026); $T = 12$, $\gamma = 2$, 1 000 draws; dashed: $c = 1$; dotted: linear, $z = 10^3$
- Nearly ridgeless ($z = 10^{-3}$): $R^2 = -1842\%$ at $c = 1$, recovering to -16.6% at $c = 1\,000$: the double descent of Figure 7 (left panel cut at -60%)
- $z = 10^3$: Sharpe ratio 0.14 at $c = 1$, 0.26 at $c = 30$, 0.08 at $c = 1\,000$ ($t = 0.5$); linear with $z = 10^3$: Sharpe 0.32 ($t = 1.7$), not beaten by RFF (information ratio -0.17)
- Reading: complexity helps up to $c \approx 30$, not up to $c = 1\,000$, with 7 of the 15 predictors and 29.5 years

Case Study: Timing Positions and Recessions



- Figure 10 of Kelly, Malamud & Zhou (2024) on course data: position $\hat{\beta}' S_t$ of RFF with $c = 1000, z = 10^3$; shaded: NBER recessions (FRED USREC); red: 6-month moving average
- Long in 72% of months, average position 0.13: a positive equity premium learned without a sign restriction
- Average position 0.07 in the 6 months before the 3 recessions and -0.01 during them, against 0.15 otherwise; the 2020 pandemic shock is not anticipated
- Three recessions are a description, not a test: the paper has 15 recessions and 90 years

Case Study: Limits and an Open Question

- **Limits** stated by the authors
 - ▶ Theory for linear ridge models on fixed random features and a single asset; a panel with cross-asset covariances is left open (Introduction, footnote 2)
 - ▶ Predictability spread evenly across features (Assumption 4); sparse LASSO-type models left to future research (footnote 22)
 - ▶ No optimal degree of nonlinearity γ ; average timing returns about half as large in 1975–2020 as in 1930–1974 (Section V.F)
- **Course data** (Seminar C3): 7 of the 15 predictors and about 30 years; the power of a two-sided 5% test for an information ratio of 0.31 is only about 39% ($t \approx 0.31\sqrt{29.5} \approx 1.7$)
- **Open question**: learned instead of random features, the embeddings of a pretrained foundation model such as Chronos-2 as S_t ; is complexity still a virtue, on a panel of course markets?

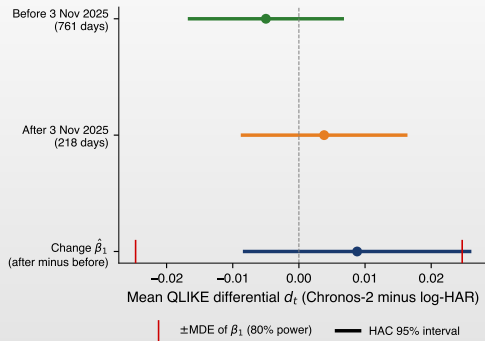
An Open Question: Skill or Memory?

- Test 2: zero-shot Chronos-2 is not distinguishable from log-HAR on SPY realised variance (QLIKE 0.231 vs 0.234, DM $p = 0.537$)
- Rahimikia, Ni & Wang (2025): on daily excess returns across global markets, off-the-shelf foundation models do poorly zero-shot and after fine-tuning; models pretrained from scratch on financial data gain
- Lopez-Lira, Tang & Zhu (2025): once a model has seen the realised values, its counterfactual forecasting skill is not identified; detected memorisation is only a lower bound (shown for LLMs; the logic applies to any pretrained forecaster)
- Open: is the parity learned structure (long memory, mean reversion) or recall of the test period, and can a short post-release window tell?
 - $d_t = L(\hat{h}_t^{FM}, RV_t) - L(\hat{h}_t^{\log HAR}, RV_t)$; memorisation implies $E[d_t | \text{post}] > E[d_t | \text{pre}]$; H_0 : equality; a rejection may also reflect a regime change

The Discovery Loop in Five Steps

- **Literature:** ask an assistant for evaluations of time-series foundation models on financial data since 2024; check every arXiv identifier and read the corpus documentation yourself
- **Hypotheses:** predictions that separate memorisation from skill: pre vs post release, never-public series, simulated series with the same dynamics
- **Data and code:** from `ch14_rv.csv`, the QLIKE differential Chronos-2 minus log-HAR, a pre/post test with HAC errors and the MDE; first reproduce the log-HAR QLIKE, 0.234
- **Robustness**, fixed in advance: a placebo release date one year earlier must show no break; simulated series cannot be memorised
- **Critique:** ask the assistant to attack “no memorisation, because the post-release test is insignificant”; expect low power and different regimes (reusable prompts in the Appendix)

Mini-Case: a Test Without Power



- Typical assistant proposal: “the parity of Chronos-2 comes from memorised data, so its edge must vanish after release”
- Mean d_t : -0.0050 before 3 Nov 2025 (761 days), +0.0038 after (218 days); change +0.0088, HAC $t = 1.00$, $p = 0.32$
- MDE = $(z_{0.975} + z_{0.8}) se_{HAC}(\hat{\beta}_1) = 0.0247$: 4.9 times the pre-release gap; with an infinite post window still 0.0169
- Reading: “no memorisation” is not a finding; power must come from the cross-section, not from waiting

 ai_discovery_case.py

Pitfalls the Researcher Must Check

- **Leakage:** pretraining overlap, full-sample scaling, HAR averages that include day t (Seminar B8 cuts QLIKE by a quarter)
 - ▶ Test: post-release window, simulated or private series; recompute the forecast for t from data truncated at $t - 1$
- **Weak benchmarks:** compare with zero (returns), log-HAR (volatility), FHS (tails), not with a random walk
- **Wrong functional:** medians used as means (mean forecast / mean RV should be near 1; Chronos-2 median: 0.79); deciles used as 1% quantiles
- **Selection:** seeds, contexts and models chosen after seeing the test; log every trial, compare the whole set with the MCS
- **Wrong inference:** DM on nested models, no multiplicity control, coverage without independence, “not significant” reported without the MDE

Your Turn: a Project Seed

- Question: does zero-shot skill for daily variance survive where memorisation is impossible: after release, on simulated series, on less-covered markets?
- Replicate first: the chapter's QLIKE of log-HAR (0.234) and Chronos-2 (0.231), DM $p = 0.537$, and the 218-day post-release window
- Deliverables
 - ▶ A pre-registered power calculation; a panel of series from the course data with a range-based variance proxy and a proxy-robust loss ([Patton \(2011\)](#))
 - ▶ Pre/post tests with placebo release dates; the MCS across models; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, leak or wrong number caught in AI_ERRORS.md, with the test that caught it ([templates](#))

When to Use What

Target	Reference model	Where deep learning can help
Daily index returns	Zero forecast	Less efficient markets (BET); many assets, many features
Realised volatility	log-HAR	Zero-shot forecasts for thousands of series without estimation
VaR 1%, ES 2.5%	FHS (GARCH filter)	Volatility filter in FHS or conformal recalibration

- Foundation models are cheap to try: no estimation, one line of code, a CPU is enough
- Their value is breadth (many series, no tuning), not accuracy on a single well-studied series

Key Takeaways

- ▣ LSTM gates can mitigate vanishing gradients (with a forget-gate bias of 0 they still vanish); attention connects distant time steps directly
- ▣ Foundation models forecast zero-shot from a context window and return quantiles
- ▣ Returns: nothing beats zero on large markets; volatility: foundation models not distinguishable from log-HAR; tails: use them only as volatility filters
- ▣ Evaluate every new model against a strong benchmark and after the release of its weights, with inference that matches the design: Clark–West for nested models, SPA or Romano–Wolf for many models, PIT, quantile scores and DQ for distributions

Check Yourself, and Next: Chapter 15

Retrieval questions

- Why does a stable RNN forget at the rate at which a VAR(1) impulse response decays?
- Why is $\exp(q_{0.5})$ of log RV a biased forecast under QLIKE?
- Why can a hybrid VaR pass Kupiec and still lose the FZ0 comparison?
- Why is “not significant after release” not evidence against memorisation?

Chapter 15: LLMs and sentiment analysis

- Can text read by a large language model forecast returns or volatility?
- The same threats return: look-ahead, memorisation of the test period, weak benchmarks
- The same tools: post-release windows, Clark–West, multiplicity control

Appendix: Norms, Spectral Radius and Explosion

- $\gamma_\varphi \|W\|_2 < 1$ is sufficient for geometric vanishing, so $\gamma_\varphi \|W\|_2 \geq 1$ is necessary for explosion (Pascanu, Mikolov & Bengio (2013))
- Spectral radius ρ : in a linear RNN, $\rho(W) > 1$ is necessary for exponential explosion ($\rho(W) = 1$ with a Jordan block still gives polynomial growth)
 - ▶ With φ the local Jacobians $D_t W$ matter: $\rho(W) = 0$ yet $\rho(DW) > 1$ is possible
- Stable VAR: impulse responses and gradients decay like $\rho(A)^k$ up to a factor k^{m-1} if A has a Jordan block of size m
- DUx_t is an innovation only if the inputs x_t are centred white noise; long-memory inputs would pass through the network

Appendix: Geometric versus Hyperbolic Decay, Fitted

- Constant gates, i.i.d. candidate input g_t , stationary start: $\text{Corr}(c_t, c_{t-k}) = f^k$; ARFIMA with $0 < d < 1/2$: $\rho_k \sim Ck^{2d-1}$
- S&P 500 absolute returns since 2000: ACF 0.289 at lag 1, 0.132 at lag 50
 - ▶ Geometric decay with its own amplitude, $\rho_k = \rho_1 f^{k-1}$ (an AR(1) signal observed with noise), matches both lags with $f = 0.984$: $b_f = 4.1$, half-life 43 days
 - ▶ That f implies 0.027 at lag 150; the data show 0.059, more than twice as much (95% band ± 0.024)
- A constant gate $f < 1$ is a short-memory filter; a long context or explicit long averages, as in HAR (Test 2), only approximate long memory over a finite range of lags

Appendix: Other Foundation Models and Their Corpora

- Chronos-Bolt: quantile levels 10%, 20%, . . . , 90% only; Chronos-2: training corpus of selected Chronos and [GIFT-Eval](#) datasets plus synthetic multivariate series; benchmarks GIFT-Eval ([Aksu et al. \(2024\)](#)) and fev-bench
- TimesFM corpus: Google Trends, Wikipedia page views, synthetic series, M4, electricity, traffic, weather; TimesFM-2.5 returns the mean plus deciles
 - ▶ Weight dates from the Hugging Face repositories: TimesFM-2.5 refreshed checkpoint (announced 15.09.2025); Chronos-2 announced 20.10.2025
- **Moirai** ([Woo et al. \(2024\)](#)): any number of variables, mixture-distribution output; Moirai 2.0 ([Liu et al. \(2025\)](#)); non-commercial licence, not used in our tests
- **Lag-Llama** ([Rasul et al. \(2023\)](#)): decoder-only model with lagged values as features and a Student- t output

Appendix: A Formal Contamination Protocol

- Pre/post regression on the loss differential $d_t = L_t^{FM} - L_t^{bench}$: $d_t = \beta_0 + \beta_1 \mathbf{1}\{t \geq \tau\} + u_t$, HAC s.e.
 - ▶ τ : release date of the weights (a common cutoff when several models are compared); memorisation predicts $\beta_1 > 0$; H_0 : $\beta_1 = 0$
 - ▶ Report the minimum detectable effect (MDE) of β_1 : with 218 post-release days it is 4.9 times the pre-release gap (AI section)
- Pre-registration: forecasts submitted before the outcome exists make contamination impossible by design
 - ▶ Live benchmarks: TS-Arena ([Meyer et al. \(2025\)](#)), Impermanent ([Garza et al. \(2026\)](#)); our post-release window is a manual version
- Ablations: removing the language-model part of LLM-based forecasters does not degrade accuracy, often improves it ([Tan et al. \(2024\)](#))
- Electricity prices: [Pan & Ezzat \(2026\)](#) benchmark on two datasets to limit contamination; foundation models are competitive but do not consistently beat domain-specific methods

Appendix: Return Forecasts, Full Table

R^2 (%) [DM p]	Historical mean	AR(1)	LSTM (5 seeds)	Chronos-2	Chronos-Bolt	TimesFM-2.5
S&P 500	+0.15 [0.095]	+0.94 [0.639]	+0.93 [0.090]	-0.26 [0.418]	-1.57 [0.078]	-3.35 [0.001]
Bitcoin	+0.07 [0.691]	-0.28 [0.326]	+0.04 [0.243]	-0.35 [0.413]	-0.61 [0.070]	-2.29 [0.001]
BET	+0.30 [0.081]	-0.00 [1.000]	+1.15 [0.014]	-0.39 [0.562]	+0.12 [0.831]	-3.27 [0.070]

- Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 -0.77% (Chronos-2), -3.42% (TimesFM-2.5)
- DM: two-sided test on squared errors against the zero forecast, Newey–West variance

Appendix: Nested Forecasts, the BET AR(1) and Rolling Windows

- BET AR(1), $R^2 = -0.00\%$ and DM $p = 1.000$, is not a degenerate forecast: s.d. 0.074%, average slope 0.046
 - ▶ Squared-error gains of -12.7 in 2020 and +14.7 in 2026 cancel; without these two years $R^2 = -0.11\%$
- Rolling or re-trained models: [Giacomini & White \(2006\)](#) test the forecasting method, estimation window included ([Diebold \(2015\)](#) on the use and abuse of DM)
- SPA refines White's Reality Check ([White \(2000\)](#)): very poor models (TimesFM) no longer dilute the test; both use a stationary bootstrap with mean block 20 days

Appendix: QLIKE, DM and MCS, Full Table

Model	QLIKE	DM vs log-HAR	p	MCS p	QLIKE after 3.11.2025
Random walk	0.326	+4.53	<0.001	0.00	0.384
HAR	0.247	+1.30	0.195	0.29	0.270
log-HAR	0.234	–	–	0.95	0.238
GARCH-t	0.283	+3.43	<0.001	0.03	0.295
LSTM	0.284	+1.68	0.094	0.29	0.233
Chronos-2	0.231	-0.62	0.537	1.00	0.242
Chronos-Bolt	0.232	-0.18	0.857	0.95	0.236
TimesFM-2.5	0.233	-0.08	0.938	0.95	0.244

- Literature: [Brini \(2026\)](#) compares nine zero-shot foundation models with eight HAR-type models on 50 assets; pooled losses favour foundation models because of a few outlier assets, while asset by asset only one small model beats log-HAR at every horizon, narrowly
- [Goel et al. \(2025\)](#): TimesFM zero-shot is a reasonable baseline; incremental fine-tuning is needed to beat econometric models

Appendix: VaR 1% Breaches and Kupiec Tests, Full Table

Breaches [Kupiec p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (26.9)	37 [0.065]	44 [0.002]	30 [0.559]	45 [0.001]	25 [0.705]	27 [0.989]
Bitcoin (28.2)	19 [0.065]	27 [0.822]	25 [0.539]	33 [0.374]	27 [0.822]	27 [0.822]
BET (26.8)	32 [0.322]	41 [0.010]	30 [0.536]	40 [0.016]	26 [0.884]	32 [0.322]

- In brackets after the market: the expected number of breaches, 1% of the days
- Kupiec: two-sided likelihood-ratio test of a 1% breach rate, χ_1^2

Appendix: COVID-19 Crash, Scaling and Overshoot

- Chronos-2 standardises the context by its mean and standard deviation, then applies $\sinh^{-1}$ ([Ansari et al. \(2025\)](#)); on 16.03.2020 the crash raised the 512-day standard deviation by 9.7% (mean absolute scale of Chronos: 3.2%, Seminar A4)
 - ▶ So the widening mixes the scale change and the reading of the crash days; only an ablation could separate them
- Not a unit error: after the crash day Chronos-2 forecast a mean $|r|$ of 12.5% for the next day, then FHS multiplies it by the 1% quantile of $r/\hat{\sigma}$ (4.7)
 - ▶ The filter extrapolates a one-day spike as a new level: VaR 4.6 times the largest loss, a high capital cost that the FZ0 loss penalises

Appendix: Berkowitz Tests and Quantile Scores, Three Markets

Market	Berkowitz p : Chronos-2	FHS	Below $q_{1\%}$ (C2)	QS: C2 / FHS	DM [p]
S&P 500	0.019	0.599	1.67%	0.510 / 0.506	+3.06 [0.002]
Bitcoin	0.595	0.306	1.17%	1.526 / 1.529	-0.90 [0.369]
BET	0.045	0.436	1.50%	0.465 / 0.460	+3.48 [<0.001]

- The Berkowitz test evaluates the linearly interpolated grid distribution function and only μ, σ , not serial independence of the PIT
- QS weights the 21 levels equally; the CRPS would weight each level by its spacing; C2: Chronos-2; tie on Bitcoin

Appendix: FZ0 Losses and MCS p -Values, Full Table

FZ0 [MCS p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500	1.309 [0.14]	1.026 [0.19]	0.999 [1.00]	1.147 [0.14]	1.084 [0.14]	1.059 [0.18]
Bitcoin	2.237 [0.34]	2.192 [0.60]	2.192 [0.60]	2.181 [0.64]	2.151 [1.00]	2.160 [0.70]
BET	1.230 [0.01]	1.001 [0.34]	0.963 [1.00]	1.032 [0.24]	1.007 [0.34]	1.027 [0.20]

- Bitcoin: the hybrid FHS with Chronos-2 volatility has the lowest FZ0 loss, but the differences are not significant
- FZ0 for (VaR 2.5%, ES 2.5%); the MCS uses the $T_{\max}$ statistic and a block bootstrap

Appendix: Conformal Recalibration, Rules and Results

After 250 calibration days:					
breaches [cc p / DQ p]		FHS	Chronos-2 raw	FHS + C2 vol.	Chronos-2 + ACI
S&P 500 (24.4)	28 [0.097 / 0.014]	42 [0.002 / <0.001]	23 [0.058 / <0.001]	21 [0.307 / 0.049]	
Bitcoin (25.7)	24 [0.449 / 0.108]	31 [0.410 / 0.026]	26 [0.087 / 0.004]	24 [0.449 / 0.191]	
BET (24.2)	26 [0.709 / 0.727]	36 [0.048 / 0.184]	23 [0.776 / 0.011]	25 [0.762 / 0.032]	
FZ0, S&P 500 [DM p]	1.013	1.142 [0.009]	1.087 [0.022]	1.205 [0.001]	

- $\hat{Q}$: the $\lceil (1 - \alpha_t)(250 + 1) \rceil$ -th smallest past score, $+\infty$ if that rank exceeds 250 or $\alpha_t \leq 0$, $-\infty$ if $\alpha_t \geq 1$; with $\gamma = 0.005$: infinite VaR on 24 (S&P 500), 100 (Bitcoin), 270 (BET) days
- Guarantee with $T = 2443$, $\gamma = 0.001$: bound 0.41, uninformative for a 1% rate; without exchangeability, weighted conformal bounds the gap by the drift ([Barber et al. \(2023\)](#))
- ES comparisons are harder still: finite-sample precision limits ([Pele & Mazurencu-Marinescu-Pele \(2026\)](#)); ES is backtestable jointly with VaR ([Nolde & Ziegel \(2017\)](#)); a GPD tail ([McNeil & Frey \(2000\)](#)) is an alternative for ES

Appendix: Case Study, Replication with Course Data

- Monthly SPY total returns from 1993 (data/market) in excess of the 3-month Treasury bill rate (FRED TB3MS)
 - ▶ Predictors of [Welch & Goyal \(2008\)](#) available: `tbl`, `lty`, `tms`, `dfy`, `infl` (FRED), `svar` from daily S&P 500 returns, the lagged excess return: 7 of the 15
 - ▶ Data limitation: the valuation ratios (`dp`, `dy`, `ep`, `de`, `b/m`), `ntis` and the bond returns `ltr`, `dfr` are not in the course data; about 30 years instead of 90
 - ▶ Method unchanged: $\gamma = 2$, $T = 12$, $P \leq 12\,000$, $\log_{10} z = -3, \dots, 3$, 1 000 draws
- Course-data results: linear kitchen sink, $z = 0^+$ ($c = 7/12$): $R^2 = -1\,540\%$, Sharpe -0.06 ; linear, $z = 10^3$: $R^2 = 0.26\%$, information ratio 0.25 ($t = 1.3$) against SPY; RFF, $c = 1\,000$: $R^2 = -6.5\%$, information ratio -0.17 ($t = -0.9$)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** "Find evaluations of time-series foundation models on financial returns or volatility since 2024: models, data, loss, benchmark, and whether the test period overlaps the pretraining corpus."
 - ▶ Semantic Scholar, Google Scholar, Elicit help with the search
- **Hypotheses:** "List predictions that separate memorisation from skill for a zero-shot volatility forecaster: pre vs post release, never-public series, simulated series with the same dynamics."
 - ▶ Diagnostics, not proof: a simulated log-HAR path tests transfer to that process; a pre/post gap can also reflect a regime change
- **Data and code:** "From ch14_rv.csv, compute the QLIKE differential Chronos-2 minus log-HAR, test pre vs post 3 November 2025 with HAC errors, and report the minimum detectable effect at 80% power."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** "Design a pre-registered test with more power: which cross-section of series, which variance proxy, which panel DM test, and how many post-release days."
- **Critique:** "Attack this claim: no memorisation, because the post-release DM test is insignificant. What is wrong with the inference?"
 - ▶ Expect: low power, absence of evidence, and different regimes: the April 2025 tariff shock lies in the pre-release sample, the post-release window is calmer
- **Report:** model versions and weight release dates, context length, mean or median, every seed and window tried

Appendix: Further Reading

- Gu, Kelly & Xiu (2020): machine learning for the cross-section of expected returns
- Kelly, Malamud & Zhou (2024): why very large models can predict returns better
- Ansari et al. (2024): Chronos, time series read as a language
- Crypto applications: liquidity forecasting Agakishiev, Härdle & Pele (2025), early warning for “zombie” crypto-assets Będowska-Sójka, Wójcik & Pele (2026), statistical classification of crypto-assets Pele et al. (2023)
- Textbooks: Franke, Härdle & Hafner (2019) and Härdle, Chen & Overbeck (2017) for the statistics of financial markets behind the benchmarks

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